

TIBU Data Quality Audit

Jenny Zhao
Wellesley College

Erez Yoeli
MIT Sloan School of Management

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1 Overview

This document presents a data quality audit (DQA) of TIBU, Kenya's national electronic TB reporting system. The audit was conducted in the context of the Keheala Study 2 randomized controlled trial (see [medRxiv:2026.02.11.26346015v1](https://medrxiv.org/content/2026.02.11.26346015v1)), but its scope is TIBU-specific and the findings stand independently of the trial results.

TIBU is a paper-to-digital reporting system: clinic staff record treatment data on the paper TB register at the point of care, and TB clerks subsequently transcribe entries from the paper register into TIBU. We therefore treat the paper register as the ground truth; a TIBU–paper discrepancy reflects a gap or error on the transcription side rather than at point of care.

The TIBU extract available for this audit covers patients registered between April 2018 and December 2019 ($N = 119,653$ nationwide), with treatment outcomes recorded through late 2021. The long outcome window reflects standard TB treatment duration: roughly six months for drug-sensitive disease and up to 18–24 months for multidrug-resistant (MDR) disease. A paper-registry audit was conducted across study clinics in multiple waves beginning in late 2018; the latest outcome dates captured by the audit fall in mid-2020, implying the final wave ran into the second half of 2020. The audit yielded 3,645 records, of which 3,370 could be matched to a corresponding TIBU record and form the basis of the DQA (see §3 on the matching procedure). Tables 1a and 1b characterize three nested cohorts: all patients registered in TIBU during the study period, patients at study clinics in TIBU, and the subset of study-clinic patients whose records were audited.

The audit covers two categories of recording error. The primary analysis (§4) compares the **treatment-outcome field** as recorded in TIBU against the paper registry, quantifying outcome misclassification. A complementary analysis of errors in the **date fields** (registration date, outcome date) is in progress and will be added as a subsequent section.

For the outcome analysis, patients recorded as transferred out in TIBU—including those assigned the MT4 code, used when drug resistance is discovered mid-treatment and the patient is referred to a drug-resistant TB program—are excluded, as a transfer is not an auditable final outcome. Records where the paper registry has no outcome are also excluded, as the audit requires ground truth. Among the remaining patients, three types of outcome discrepancy are distinguished:

- **False positive** in the digital record (a Type I error): TIBU records a successful outcome (treatment completion or cure) but the paper registry records a failure (death, treatment failure, or loss to follow-up). Clinically: TIBU *overstates* the program's success.
- **False negative** in the digital record (a Type II error): TIBU records a failure while the paper registry records success. Clinically: TIBU *understates* the program's success.
- **Missing in TIBU**: TIBU contains no outcome but the paper registry does. This is distinct from loss-to-follow-up, which is itself a recorded outcome: “missing in TIBU” means the clinic did record an outcome on paper, but the TIBU entry was never created or never finalized.

The cross-tabulation in Table 2 maps the full distribution of these discrepancies across outcome categories; subsequent tables stratify error rates by clinic characteristics (province, urbanicity, and size) and patient characteristics (sex, age, disease type, and HIV status); and the smoothed time-series

figures trace how error rates vary across the study period and with the age of the patient record.

Key findings. Across 3,054 classifiable audit records: false-positive raw rate 1.0%; false-negative rate 0.7%; records missing from TIBU 3.8%. The false-positive rate likely overstates true TIBU misclassification: $\sim 70\%$ of false-positive records appear to be audit-snapshot artifacts (paper was updated after the audit transcribed it; TIBU then reflected the revised outcome), and a date-direction sensitivity classification (§5) cuts the rate roughly in half (to 0.5%). False-negative and missing-in-TIBU rates are not affected by this caveat. The missing-in-TIBU rate is dominated by a COVID-era reporting gap ($\sim 15\%$ for treatment outcomes in 2020 Q1–Q2), concentrated at two clinics; outside that window, all three outcome-error rates are low and roughly flat across the study period.

2 Patient Characteristics

Tables 1a and 1b describe the study population across three cohorts: all patients registered in the TIBU electronic system during the study period, patients at study clinics recorded in TIBU, and the subset of study-clinic patients whose records were audited in the paper registry DQA.

Table 1a: Individual and disease characteristics across TIBU and paper registry cohorts.

Characteristic	Digital				Paper	
	All		Study clinics		Study clinics	
	(N=119,653)		(N=14,962)		(N=3,263)	
	N	(%)	N	(%)	N	(%)
<i>Individual characteristics</i>						
Male	75,597	(63.2%)	9,514	(63.6%)	2,037	(64.5%)
Mean age (years)	36.1		34.4		32.7	
<i>Disease characteristics</i>						
Bacteriologically confirmed	62,149	(51.9%)	11,243	(75.1%)	2,223	(70.4%)
Drug resistant	1,343	(1.1%)	2	(0.0%)	3	(0.1%)
Extrapulmonary TB	17,558	(14.7%)	1,794	(12.0%)	360	(11.4%)
HIV coinfection	32,077	(26.8%)	3,981	(26.9%)	717	(23.2%)
Retreatment	8,351	(7.0%)	1,262	(8.7%)	290	(9.4%)

Table 1b: Treatment outcomes across TIBU and paper registry cohorts.

Characteristic	Digital				Paper	
	All		Study clinics		Study clinics	
	(N=119,653)		(N=14,962)		(N=3,263)	
	N	(%)	N	(%)	N	(%)
<i>Outcomes</i>						
Cured	25,399	(21.2%)	6,757	(45.2%)	1,114	(34.1%)
Treatment completed	31,719	(26.5%)	6,637	(44.4%)	1,468	(45.0%)
Died	4,911	(4.1%)	765	(5.1%)	181	(5.5%)
Treatment failed	382	(0.3%)	123	(0.8%)	34	(1.0%)
Lost to follow-up	3,594	(3.0%)	680	(4.5%)	125	(3.8%)
Not completed	47,184	(39.4%)	0	(0.0%)	87	(2.7%)
Transfer out	6,382	(5.3%)	0	(0.0%)	138	(4.2%)
Not TB	0	(0.0%)	0	(0.0%)	0	(0.0%)
Missing	82	(0.1%)	0	(0.0%)	116	(3.6%)

3 Record Matching Procedure

Placeholder — details of the matching procedure (identifiers used, handling of near-duplicates, reasons for non-matches) are currently being confirmed with the Keheala data team and will be added in a subsequent revision. The headline figures are: 3,645 audited paper records, of which 3,370 (92.5%) were matched to a TIBU entry; the remaining 275 unmatched audit records are excluded from the DQA because no corresponding TIBU counterpart exists.

4 Outcome Errors

4.1 Crosstab of TIBU vs. Paper Registry Outcomes

Table 2 shows the full cross-tabulation of treatment outcomes recorded in TIBU against outcomes recorded in the paper registry for all matched patients ($N = 3,370$). Counts on the diagonal are concordant; off-diagonal cells are the discrepancies that drive the error rates reported in the subsequent tables. The shaded regions mark the false-positive (red), false-negative (green), and missing-in-TIBU (blue) partitions used throughout the document.

Table 2: Cross-tabulation of treatment outcomes in TIBU vs. paper registry.

Outcome in TIBU	Treatment Outcome in Paper Registry							Total					
	Blank		C		D		F		LTFU		TC		
	<i>N</i>	(%)	<i>N</i>	(%)	<i>N</i>	(%)	<i>N</i>		(%)	<i>N</i>	(%)	<i>N</i>	(%)
Blank	80	(69.0%)	21	(18.1%)	1	(0.9%)	1	(0.9%)	2	(1.7%)	11	(9.5%)	116
C	41	(3.7%)	1,008	(90.5%)	1	(0.1%)	3	(0.3%)	7	(0.6%)	54	(4.8%)	1,114
D	4	(2.2%)	2	(1.1%)	166	(91.7%)	0	(0.0%)	1	(0.6%)	8	(4.4%)	181
F	0	(0.0%)	1	(2.9%)	1	(2.9%)	25	(73.5%)	5	(14.7%)	2	(5.9%)	34
LTFU	3	(2.4%)	2	(1.6%)	1	(0.8%)	0	(0.0%)	113	(90.4%)	6	(4.8%)	125
NC	10	(11.5%)	34	(39.1%)	6	(6.9%)	0	(0.0%)	4	(4.6%)	32	(36.8%)	87
TC	33	(2.2%)	76	(5.2%)	7	(0.5%)	0	(0.0%)	13	(0.9%)	1,339	(91.2%)	1,468
TO	6	(4.3%)	45	(32.6%)	5	(3.6%)	4	(2.9%)	14	(10.1%)	63	(45.7%)	138
Total	177		1,189		188		33		159		1,515		3,263

Note: Percentages are row shares.

Red = False positive (TIBU: success; paper: failure).

Green = False negative (TIBU: failure; paper: success).

Blue = Missing in TIBU (TIBU: no outcome; paper: outcome present).

4.2 Error Rates by Clinic Characteristics

Table 3 reports false-positive, false-negative, and missing-in-TIBU rates stratified by province, urbanicity, and clinic size.

Table 3: DQA error rates by province, urbanicity, and clinic size.

	<i>N</i> (%)		False positive <i>N</i> (%)		False negative <i>N</i> (%)		Missing in TIBU <i>N</i> (%)	
<i>By province</i>								
Central	24	(0.8%)	0	(0.0%)	0	(0.0%)	11	(45.8%)
Coast	686	(23.2%)	7	(1.0%)	2	(0.3%)	12	(1.7%)
Eastern North	2	(0.1%)	0	(0.0%)	0	(0.0%)	0	(0.0%)
Eastern South	254	(8.6%)	4	(1.6%)	2	(0.8%)	22	(8.7%)
Nairobi	362	(12.3%)	5	(1.4%)	4	(1.1%)	3	(0.8%)
North Eastern	724	(24.5%)	9	(1.2%)	3	(0.4%)	2	(0.3%)
Nyanza North	382	(12.9%)	2	(0.5%)	7	(1.8%)	3	(0.8%)
Nyanza South	41	(1.4%)	1	(2.4%)	0	(0.0%)	0	(0.0%)
Rift Valley North	38	(1.3%)	0	(0.0%)	0	(0.0%)	0	(0.0%)
Rift Valley South	379	(12.8%)	2	(0.5%)	2	(0.5%)	29	(7.7%)
Western	32	(1.1%)	1	(3.1%)	1	(3.1%)	1	(3.1%)
<i>By urban/rural</i>								
Urban	928	(31.4%)	10	(1.1%)	5	(0.5%)	18	(1.9%)
Rural	1,992	(67.5%)	21	(1.1%)	16	(0.8%)	65	(3.3%)
<i>By clinic size</i>								
Large clinic	1,174	(39.8%)	17	(1.4%)	7	(0.6%)	22	(1.9%)
Small clinic	1,779	(60.2%)	14	(0.8%)	14	(0.8%)	90	(5.1%)
All	2,953	(100.0%)	31	(1.0%)	21	(0.7%)	112	(3.8%)

Note: Urban/rural classification is based on clinic records. Large clinics are those with at least 6 patients registered in TIBU (median); small clinics are those with fewer.

4.3 Error Rates by Patient Characteristics

Table 4 reports false-positive, false-negative, and missing-in-TIBU rates broken down by individual characteristics and disease characteristics.

Table 4: DQA error rates by individual and disease characteristics.

	<i>N</i>	(%)	False positive <i>N</i>	(%)	False negative <i>N</i>	(%)	Missing in TIBU <i>N</i>	(%)
<i>Individual characteristics</i>								
Male (%)	1,872	(63.4%)	19	(1.0%)	14	(0.7%)	68	(3.6%)
<i>Age group</i>								
<15	420	(14.2%)	1	(0.2%)	3	(0.7%)	5	(1.2%)
15–34	1,334	(45.2%)	18	(1.3%)	11	(0.8%)	35	(2.6%)
35–54	866	(29.3%)	6	(0.7%)	5	(0.6%)	29	(3.3%)
55+	297	(10.1%)	6	(2.0%)	2	(0.7%)	8	(2.7%)
<i>Disease characteristics</i>								
Bacteriologically confirmed	2,041	(69.1%)	25	(1.2%)	14	(0.7%)	61	(3.0%)
Extrapulmonary	331	(11.2%)	2	(0.6%)	0	(0.0%)	8	(2.4%)
Retreatment	275	(9.3%)	5	(1.8%)	1	(0.4%)	5	(1.8%)
HIV positive	667	(22.6%)	6	(0.9%)	8	(1.2%)	20	(3.0%)
All	2,953	(100.0%)	31	(1.0%)	21	(0.7%)	112	(3.8%)

4.4 Error Rates Over Time

The TIBU dataset used for these analyses reflects a single final extract. The latest treatment-outcome dates cluster in late December 2021, and we treat the snapshot as having occurred on 1 January 2022. Figure 2 plots DQA error rates against outcome date, estimated via locally-weighted regression (lowess); shaded bands show percentile bootstrap 95% confidence intervals (500 resamples). For records that are missing in TIBU (and therefore have no TIBU outcome date), the paper-registry outcome date is used in its place, so that these records remain representable on the time axis.

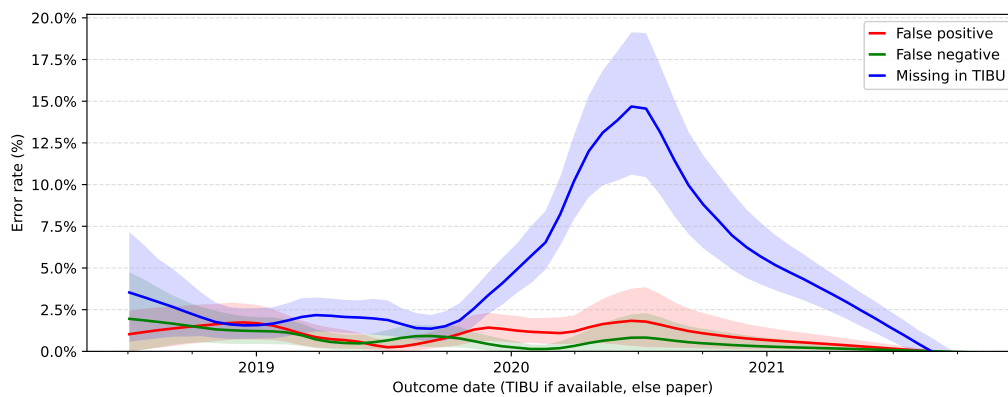


Figure 2: DQA error rates over time by outcome date (TIBU when available, paper otherwise). Lowess smooth with percentile bootstrap 95% CI bands.

A sensitivity check using paper-registry outcome dates throughout (rather than TIBU date with paper fallback for missing-in-TIBU records) is reported in Appendix A. The COVID-era spike in missing-in-TIBU remains clearly visible there, confirming that it is driven by when outcomes actually occurred (as captured on paper), not by when TIBU happened to record them.

The dominant feature of the figure is a pronounced peak in the missing-in-TIBU rate (to roughly 15%) for records with outcome dates in 2020 Q1–Q2, coinciding with the onset of the COVID-19 pandemic response in Kenya. Of the 116 missing-in-TIBU records in the classifiable sample, 63 fall in these two quarters, and 50 are concentrated at just two clinics whose overall missing-in-TIBU rates (12.7% and 15.4%) are an order of magnitude above the study-wide rate of 3.8%. At these clinics, paper-registry entries continued through the pandemic but TIBU outcomes were largely not recorded. NC is a TIBU interim code denoting that treatment is still in progress; treating it as equivalent to blank (rather than as a medical failure) rolls the affected records into the missing-in-TIBU category, which accounts for 76 of the 116 (the remaining 40 have TIBU entirely blank). False-positive and false-negative rates are low and roughly flat across the study period. Equivalent breakdowns under the alternative “date-gated” definition of error introduced in §5 are collected in Appendix B.

5 Incorporating Date Fields to Assess Outcome Errors

Up to this point we have treated the paper register as ground truth. But the audit itself can err: through transcription mistakes, by matching a paper record to the wrong patient, or by transcribing an out-of-date entry for the right patient. The outcome-date comparison in this section speaks directly to the last of these failure modes.

Although an audit of the date fields was not a priority of the trial’s data-quality programme, the audit team transcribed dates alongside outcomes for 3,186 of the 3,645 audit records (87.4%). A comparison of outcome dates from TIBU and the paper audit can provide insight on the nature of the inconsistency between TIBU and the paper records as digitized in the audit.

For DQA-matched records where both the TIBU and paper outcome dates are parseable, we can compare the two directly (Table 5 and Figure 3).

Table 5: TIBU–paper outcome-date agreement, by outcome-error category.

Category	N	TIBU – paper (days)					TIBU date is	
		≤ 7 days	8–30 days	31–90 days	91–180 days	>180 days	Earlier	Later
No error	2,736	2123 (77.6%)	203 (7.4%)	130 (4.8%)	51 (1.9%)	229 (8.4%)	365 (13.3%)	441 (16.1%)
False positive	23	3 (13.0%)	3 (13.0%)	4 (17.4%)	10 (43.5%)	3 (13.0%)	4 (17.4%)	16 (69.6%)
False negative	18	2 (11.1%)	3 (16.7%)	5 (27.8%)	5 (27.8%)	3 (16.7%)	13 (72.2%)	4 (22.2%)

Note: Denominator is records where both the TIBU and paper outcome dates are non-blank and parseable. Missing-in-TIBU records are excluded (TIBU date is blank by construction). “TIBU earlier” and “TIBU later” together sum to < 100% because some records agree exactly.

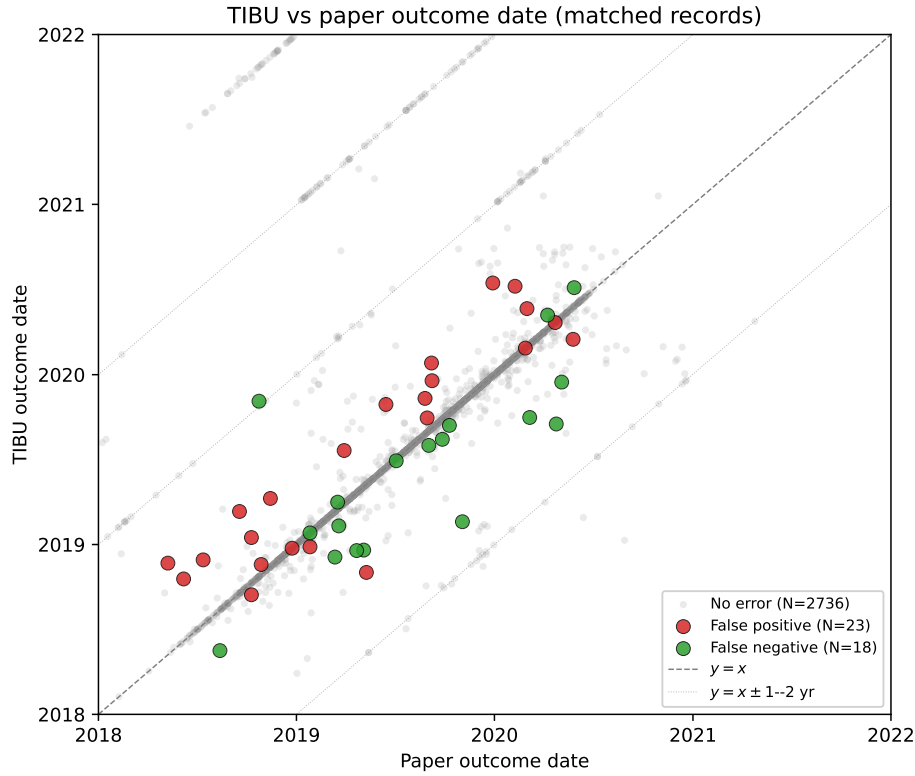


Figure 3: TIBU outcome date vs. paper outcome date for DQA-matched records with parseable dates on both sides (axes clipped to the 2018–2022 study window). Points on the $y = x$ dashed line agree exactly. False-positive points (red) sit predominantly above the line (TIBU later), false-negative points (green) predominantly below (TIBU earlier). The multiple parallel 45-degree lines arise because it is common for TIBU and the paper audit to disagree on the year of the record (without changing the recorded outcome).

Among no-error records the two dates agree to within a week $\sim 78\%$ of the time, with a small symmetric tail. The pattern is qualitatively different for the two error classes: for false-positive records, TIBU is *later* than the audit-transcribed paper date in 16 of 23 cases ($\sim 70\%$); for false-negative records, TIBU is *earlier* in 13 of 18 ($\sim 72\%$). This suggests that the majority of apparent false positives are not in fact errors, but result from the paper audit occasionally having employed less up-to-date records than TIBU.

Operationalizing this reading, we define a *date-gated* sensitivity classification: a TIBU–paper outcome disagreement is counted as an error only when the paper outcome date is strictly later than the TIBU outcome date. When TIBU is the more recent reading, the record is treated as concordant on the assumption that TIBU reflects a post-audit paper update. Missing-in-TIBU is unchanged because TIBU is blank by construction. Table 6 shows the resulting headline-vs-gated comparison; under the gate, the false-positive rate falls by roughly half (from 1.05% to 0.51%), the false-negative rate falls only slightly (from 0.71% to 0.58%), and the missing-in-TIBU rate is unchanged (3.79%). Equivalent breakdowns of Tables 3 and 4 and Figure 2 under the date-gated definition are collected in Appendix B.

Table 6: Headline vs. date-gated DQA error counts and rates.

Error type	Headline		Date-gated	
	N	Rate	N	Rate
False positive	31	1.05%	15	0.51%
False negative	21	0.71%	17	0.58%
Missing in TIBU	112	3.79%	112	3.79%

Note: Denominator is the $N = 2,953$ classifiable matched records. Under the date-gated definition, a TIBU–paper outcome disagreement is counted as an error only when the paper outcome date is strictly later than the TIBU outcome date; otherwise the record is treated as concordant on the assumption that TIBU reflects a post-audit paper update. The rule cannot be applied to records with an unparseable date on either side ($n = 8$ false-positive and $n = 3$ false-negative records), and such records preserve their headline classification. Missing-in-TIBU is unaffected since TIBU is blank by construction.

Appendix

A Sensitivity check: paper-registry outcome dates

Figure 4 replays the over-time analysis of §4 using the paper-registry outcome date throughout, rather than TIBU's outcome date with paper fallback for missing-in-TIBU records. Records with no parseable paper date are excluded ($\sim 13\%$ of the classifiable sample). The COVID-era peak in missing-in-TIBU is unchanged in shape and magnitude, indicating that it is driven by when outcomes actually occurred (as captured on paper) and not by when TIBU happened to record them.

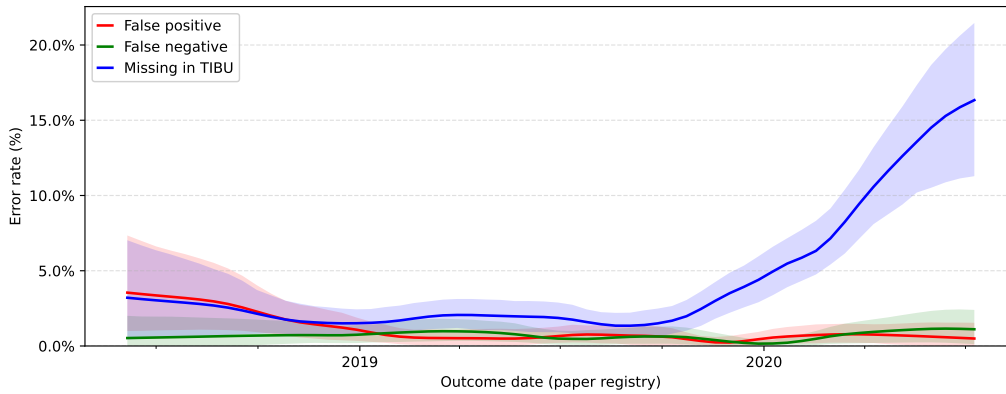


Figure 4: DQA error rates over time using paper-registry outcome dates throughout. Records without a parseable paper date are excluded. Compare with Figure 2.

B Error rates under the date-gated definition

This appendix recreates the breakdowns of Tables 3 and 4 and Figure 2 under the date-gated classification introduced in §5. A matched record is counted as a TIBU error only if the TIBU and paper outcomes disagree *and* the paper outcome date is strictly later than the TIBU outcome date; otherwise, on the reading that TIBU may reflect a post-audit paper update, the record is treated as concordant. Missing-in-TIBU is unchanged because TIBU is blank by construction. Records with an unparseable date on either side preserve their headline classification ($n = 8$ false-positive and $n = 3$ false-negative records). Headline-vs-gated marginal totals are in Table 6.

Table 7: DQA error rates by province, urbanicity, and clinic size, under the date-gated definition. Compare with Table 3.

	<i>N</i> (%)		False positive <i>N</i> (%)		False negative <i>N</i> (%)		Missing in TIBU <i>N</i> (%)	
<i>By province</i>								
Central	24	(0.8%)	0	(0.0%)	0	(0.0%)	11	(45.8%)
Coast	686	(23.2%)	3	(0.4%)	2	(0.3%)	12	(1.7%)
Eastern North	2	(0.1%)	0	(0.0%)	0	(0.0%)	0	(0.0%)
Eastern South	254	(8.6%)	3	(1.2%)	2	(0.8%)	22	(8.7%)
Nairobi	362	(12.3%)	2	(0.6%)	4	(1.1%)	3	(0.8%)
North Eastern	724	(24.5%)	4	(0.6%)	2	(0.3%)	2	(0.3%)
Nyanza North	382	(12.9%)	0	(0.0%)	5	(1.3%)	3	(0.8%)
Nyanza South	41	(1.4%)	1	(2.4%)	0	(0.0%)	0	(0.0%)
Rift Valley North	38	(1.3%)	0	(0.0%)	0	(0.0%)	0	(0.0%)
Rift Valley South	379	(12.8%)	1	(0.3%)	2	(0.5%)	29	(7.7%)
Western	32	(1.1%)	1	(3.1%)	0	(0.0%)	1	(3.1%)
<i>By urban/rural</i>								
Urban	928	(31.4%)	3	(0.3%)	5	(0.5%)	18	(1.9%)
Rural	1,992	(67.5%)	12	(0.6%)	12	(0.6%)	65	(3.3%)
<i>By clinic size</i>								
Large clinic	1,174	(39.8%)	11	(0.9%)	5	(0.4%)	22	(1.9%)
Small clinic	1,779	(60.2%)	4	(0.2%)	12	(0.7%)	90	(5.1%)
All	2,953	(100.0%)	15	(0.5%)	17	(0.6%)	112	(3.8%)

Note: Urban/rural classification is based on clinic records. Large clinics are those with at least 6 patients registered in TIBU (median); small clinics are those with fewer.

Table 8: DQA error rates by individual and disease characteristics, under the date-gated definition. Compare with Table 4.

	<i>N</i> (%)		False positive <i>N</i> (%)		False negative <i>N</i> (%)		Missing in TIBU <i>N</i> (%)	
<i>Individual characteristics</i>								
Male (%)	1,872	(63.4%)	10	(0.5%)	12	(0.6%)	68	(3.6%)
<i>Age group</i>								
<15	420	(14.2%)	1	(0.2%)	3	(0.7%)	5	(1.2%)
15–34	1,334	(45.2%)	9	(0.7%)	8	(0.6%)	35	(2.6%)
35–54	866	(29.3%)	3	(0.3%)	4	(0.5%)	29	(3.3%)
55+	297	(10.1%)	2	(0.7%)	2	(0.7%)	8	(2.7%)
<i>Disease characteristics</i>								
Bacteriologically confirmed	2,041	(69.1%)	12	(0.6%)	10	(0.5%)	61	(3.0%)
Extrapulmonary	331	(11.2%)	1	(0.3%)	0	(0.0%)	8	(2.4%)
Retreatment	275	(9.3%)	2	(0.7%)	1	(0.4%)	5	(1.8%)
HIV positive	667	(22.6%)	6	(0.9%)	7	(1.0%)	20	(3.0%)
All	2,953	(100.0%)	15	(0.5%)	17	(0.6%)	112	(3.8%)

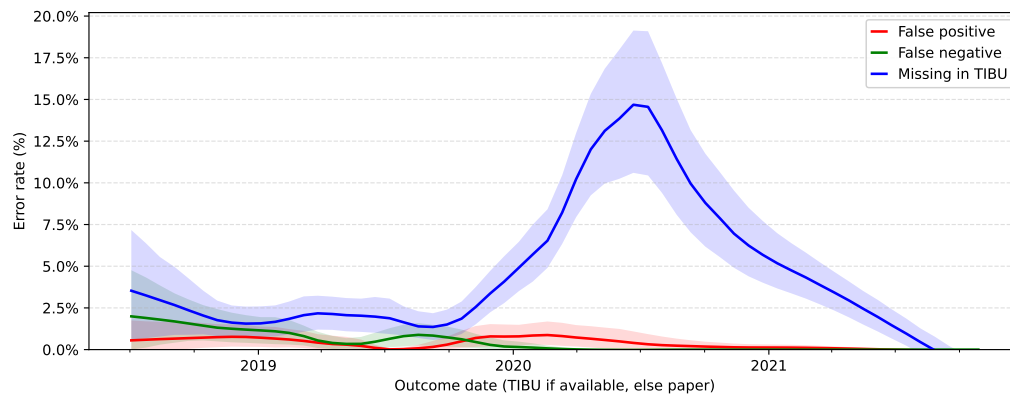


Figure 5: DQA error rates over time, under the date-gated definition. Compare with Figure 2.